

SOFTWARE REQUIREMENTS SPECIFICATION

Project: A Snake Game

A Modular Snake Game with Multiple Game Modes, Power-Ups, Survival Challenges, Multiplayer and Leaderboards

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Revision History

Version	Date	Author	Change Summary	Approval
0.1	09-09-2026	Thejaswi S, Vishnu Maddirala, V S Shashidharan, Vishwas Hosamath	Initial draft SRS created from the course SRS template	Pending

Approvals

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, security, interface, and verification requirements for A Snake Game. The system is designed as a modular Snake gaming platform providing Classic, Obstacle, Power-Up, Two-Player Battle, and Survival modes.

This document serves as a reference for developers, testers, project evaluators, and other stakeholders and establishes measurable criteria for verifying that the implemented system satisfies its requirements.

1.2 Scope

The system provides an interactive Snake gaming experience in which players control a snake within a bounded game board and attempt to achieve the highest possible score while avoiding collisions.

- Classic Mode
- Obstacle Mode
- Power-Up Mode
- Two-Player Battle Mode
- Survival Mode
- User registration and authentication
- Player profiles
- Multiple difficulty levels
- Score calculation and storage
- Personal best scores and leaderboards
- Game statistics and settings
- Administrative functionality
- Security controls and error handling

The project focuses on the software implementation of the game and its supporting services. It does not include physical gaming hardware, commercial payment systems, or external gaming services.

1.3 Audience

- Software developers
- Software testers
- Project team members
- Course faculty
- Project evaluators
- System administrators
- End users/players

1.4 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
FR	Functional Requirement
NFR	Non-Functional Requirement
RTM	Requirements Traceability Matrix
UML	Unified Modeling Language
FPS	Frames Per Second
Admin	System Administrator
Player	A user who plays the game
Game Session	A single instance of gameplay
Power-Up	A temporary item providing a special ability or effect
Leaderboard	A ranked list of players based on scores
Collision	Contact between the snake and an invalid game object

2. Overall Description

2.1 Product Perspective

A Snake Game is a standalone software application consisting of a graphical user interface, game engine, game mode manager, scoring system, user management system, database, and administrative components. The game engine handles real-time gameplay mechanics such as snake movement, food generation, collision detection, scoring, and game-state management. Supporting modules provide authentication, profile management, score storage, leaderboard management, and configuration.

The system shall use a modular architecture so that individual game modes and supporting components can be developed, tested, and maintained independently.

2.2 Major Product Functions

- Register and authenticate players
- Manage player profiles
- Start, pause, resume, restart and end games
- Select game mode and difficulty
- Control snake movement
- Generate and consume food
- Detect collisions
- Calculate and store scores
- Provide Classic, Obstacle, Power-Up, Two-Player Battle and Survival modes
- Manage power-ups and progressive difficulty
- Maintain personal bests and leaderboards
- Display game statistics
- Provide authorized administration functions

2.3 User Roles and Characteristics

Player: The primary user. The player should have basic computer knowledge and be able to navigate menus, select game modes, control the snake, understand scores, view leaderboards, and configure basic settings.

Administrator: An authorized user responsible for permitted system management tasks such as viewing users and statistics, managing accounts, reviewing leaderboard data, and configuring selected parameters.

2.4 Operating Environment

- Windows-based desktop/laptop systems
- Keyboard input
- Display resolution of at least 1280 × 720
- Local database or configured application database
- Supported modern runtime/browser environment depending on implementation technology

2.5 Constraints

- The game shall remain playable using standard keyboard controls.
- The system shall provide consistent game behavior across supported environments.
- Game-state transitions shall be handled consistently.
- Player scores shall not be modifiable through normal player interactions.
- The system shall use modular components to simplify maintenance.
- The project shall be implementable within the academic project timeline.

3. External Interface Requirements

3.1 User Interface

- Login/Register screen
- Main menu
- Game mode selection
- Difficulty selection
- Game board
- Score display
- Pause menu
- Game-over screen
- Leaderboard
- Player profile
- Settings
- Administrator interface

During gameplay, the interface shall display the current score, game mode, difficulty, snake, food, and applicable obstacles or power-ups.

3.2 Hardware Interfaces

The system shall primarily use a keyboard for player input and a display/monitor for graphical output. In Two-Player Battle Mode, Player 1 shall use W/A/S/D and Player 2 shall use the Arrow Keys by default.

3.3 Software Interfaces

- User Interface Module
- Game Engine
- Game Mode Manager
- Collision Detection Module
- Score Manager
- Power-Up Manager
- Difficulty Manager

- User Management Module
- Leaderboard Module
- Database Layer
- Administration Module

3.4 Communication Interfaces

If a client-server or web-based implementation is used, communication between application components shall use appropriate application protocols and secure transport for authenticated network communication. If implemented as a standalone local application, network communication shall not be required for core gameplay.

4. System Features (Detailed)

Each functional requirement is assigned a unique identifier using the format SNAKE-F-###.

Req ID	Feature	Requirement	Priority	Test Case
SNAKE-F-001	Player Registration	The system shall allow a new player to create an account using a unique username and password.	High	TC-AUTH-001
SNAKE-F-002	Player Authentication	The system shall authenticate registered players using their username and password.	High	TC-AUTH-002
SNAKE-F-003	Player Profile	The system shall allow authenticated players to view and update permitted profile information.	Medium	TC-PROFILE-001
SNAKE-F-004	Logout	The system shall allow authenticated users to securely log out.	Medium	TC-AUTH-003
SNAKE-F-005	Start Game	The system shall allow a player to start a new game after selecting a valid game mode and difficulty.	High	TC-GAME-001
SNAKE-F-006	Game Mode Selection	The system shall allow the player to select Classic, Obstacle, Power-Up, Two-Player Battle, or Survival mode.	High	TC-MODE-001
SNAKE-F-007	Difficulty Selection	The system shall allow the player to select an available difficulty level before starting an	Medium	TC-DIFF-001

		applicable game.		
SNAKE-F-008	Pause and Resume	The system shall allow a player to pause and resume a game during gameplay.	Medium	TC-GAME-002
SNAKE-F-009	Snake Movement	The system shall move the snake continuously according to player input and configured game speed.	High	TC-MOVE-001
SNAKE-F-010	Food Consumption	The system shall detect when the snake consumes food and shall increase snake length and score accordingly.	High	TC-FOOD-001
SNAKE-F-011	Food Generation	The system shall generate food at a valid unoccupied position on the game board.	High	TC-FOOD-002
SNAKE-F-012	Collision Detection	The system shall detect collisions between the snake and walls, its body, obstacles, or an opponent snake where applicable.	High	TC-COLLISION-001
SNAKE-F-013	Score Calculation	The system shall calculate the player's score based on applicable game rules and gameplay events.	High	TC-SCORE-001
SNAKE-F-014	Obstacle Mode	The system shall provide an Obstacle Mode containing obstacles that the snake must avoid.	High	TC-OBSTACLE-001
SNAKE-F-015	Power-Up Mode	The system shall provide Power-Up Mode in which temporary power-ups appear during gameplay.	High	TC-POWERUP-001
SNAKE-F-016	Two-Player Battle	The system shall allow two players to control separate snakes simultaneously on the same game board.	High	TC-MULTI-001

SNAKE-F-017	Survival Mode	The system shall provide a Survival Mode in which gameplay difficulty progressively increases as the player survives.	High	TC-SURVIVAL-001
SNAKE-F-018	Classic Mode	The system shall provide a Classic Mode in which a single player collects food, grows the snake, and attempts to achieve the highest score without invalid collision.	High	TC-CLASSIC-001
SNAKE-F-019	Save Game Score	The system shall save the final score and relevant game information when an authenticated player's game ends.	High	TC-SCORE-002
SNAKE-F-020	Leaderboard	The system shall display a leaderboard containing ranked player scores.	Medium	TC-LEADER-001

4.1 Game Mode Details

Classic Mode

The traditional Snake experience. The snake collects food, grows in length, increases score, and ends the game when an applicable collision occurs.

Obstacle Mode

The board contains obstacles that the snake must avoid. Collision with an obstacle results in Game Over.

Power-Up Mode

Temporary power-ups may include Speed Boost, Slow Down, Shield, Score Multiplier, and Snake Shrink. Effects are applied for configured durations.

Two-Player Battle Mode

Two players control separate snakes on the same board using separate keyboard controls. Food collection, scoring, and snake-to-snake collision are supported.

Survival Mode

The objective is to survive as long as possible while difficulty progressively increases through changes such as increased speed, additional obstacles, or reduced safe movement area.

5. Non-Functional Requirements

Req ID	Requirement	Category	Priority	Acceptance Criteria
SNAKE-NF-001	The system shall	Performance	High	Measured response

	process valid player movement input within 100 ms under normal operating conditions.			≤100 ms
SNAKE-NF-002	The game shall maintain at least 30 FPS during normal single-player gameplay on the supported hardware configuration.	Performance	High	FPS measurement ≥30
SNAKE-NF-003	The system shall preserve successfully stored player scores without unintended modification or loss under normal operating conditions.	Reliability	High	Repeated retrieval matches stored score
SNAKE-NF-004	A new player shall be able to start a Classic Mode game within three primary menu interactions after successful login.	Usability	Medium	Usability test passes
SNAKE-NF-005	The system shall separate major game components into modular components to facilitate independent testing and maintenance.	Maintainability	High	Architecture/code review confirms modular separation
SNAKE-NF-006	The system shall provide consistent gameplay behavior across the supported operating environment.	Compatibility	Medium	Compatibility test passes

5.1 Security

5.1.1 Security Objectives

SNAKE-SO-001 — Account Protection: The system shall protect player and administrator accounts from unauthorized access through authentication and appropriate credential handling.

SNAKE-SO-002 — Data Integrity: The system shall protect player scores, leaderboard information, and other persistent game data from unauthorized modification.

5.1.2 Security Requirements

Req ID	Requirement	Priority	Test Case
SNAKE-SR-001	The system shall store	High	TC-SEC-001

	passwords using a secure one-way password hashing mechanism and shall not store plaintext passwords.		
SNAKE-SR-002	The system shall restrict administrator functions to authenticated and authorized administrator accounts.	High	TC-SEC-002
SNAKE-SR-003	The system shall validate user-provided input before processing or storing it.	High	TC-SEC-003
SNAKE-SR-004	The system shall prevent normal player interfaces from directly modifying stored scores or leaderboard records.	High	TC-SEC-004
SNAKE-SR-005	The system shall terminate authenticated sessions according to the configured session-management policy and prevent access to protected functionality after logout.	Medium	TC-SEC-005

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

Attribute	Expected Quality
Performance	Responsive controls and smooth gameplay.
Reliability	Consistent operation without unexpected crashes during normal use.
Usability	Clear interface understandable to new players.
Maintainability	Modular components that can be modified independently.
Security	Protection of accounts and persistent game information.
Compatibility	Correct operation in supported environments.

6.2 Acceptance Criteria

- All high-priority functional requirements are implemented and verified.
- All five game modes operate according to their defined rules.
- Movement and collision detection function correctly.
- Scores are calculated and stored correctly.
- Leaderboards display correct ranking information.
- Authentication and authorization requirements pass testing.
- Required security requirements pass their associated tests.
- No critical defects remain in the final build.
- Required non-functional requirements meet their measurable targets.

- The RTM demonstrates coverage of requirements by design and testing.

6.3 Acceptance Test Suite

Test Suite	Purpose
Authentication Tests	Verify registration, login and logout
Game Management Tests	Verify game initialization and state controls
Movement Tests	Verify snake controls and movement
Food Tests	Verify food generation and consumption
Collision Tests	Verify wall, self, obstacle and player collision
Game Mode Tests	Verify all five game modes
Power-Up Tests	Verify power-up effects and expiration
Multiplayer Tests	Verify Two-Player Battle
Survival Tests	Verify progressive difficulty
Score Tests	Verify score calculation and storage
Leaderboard Tests	Verify ranking
Security Tests	Verify authentication, authorization and data protection
Performance Tests	Verify response time and FPS
Usability Tests	Verify ease of navigation
Compatibility Tests	Verify supported environments

7. UML Use-Case Diagrams

This section presents the two required UML Use-Case diagrams for A Snake Game, covering the Player and Administrator perspectives.

7.1 Player Use-Case Diagram

Actor: Player

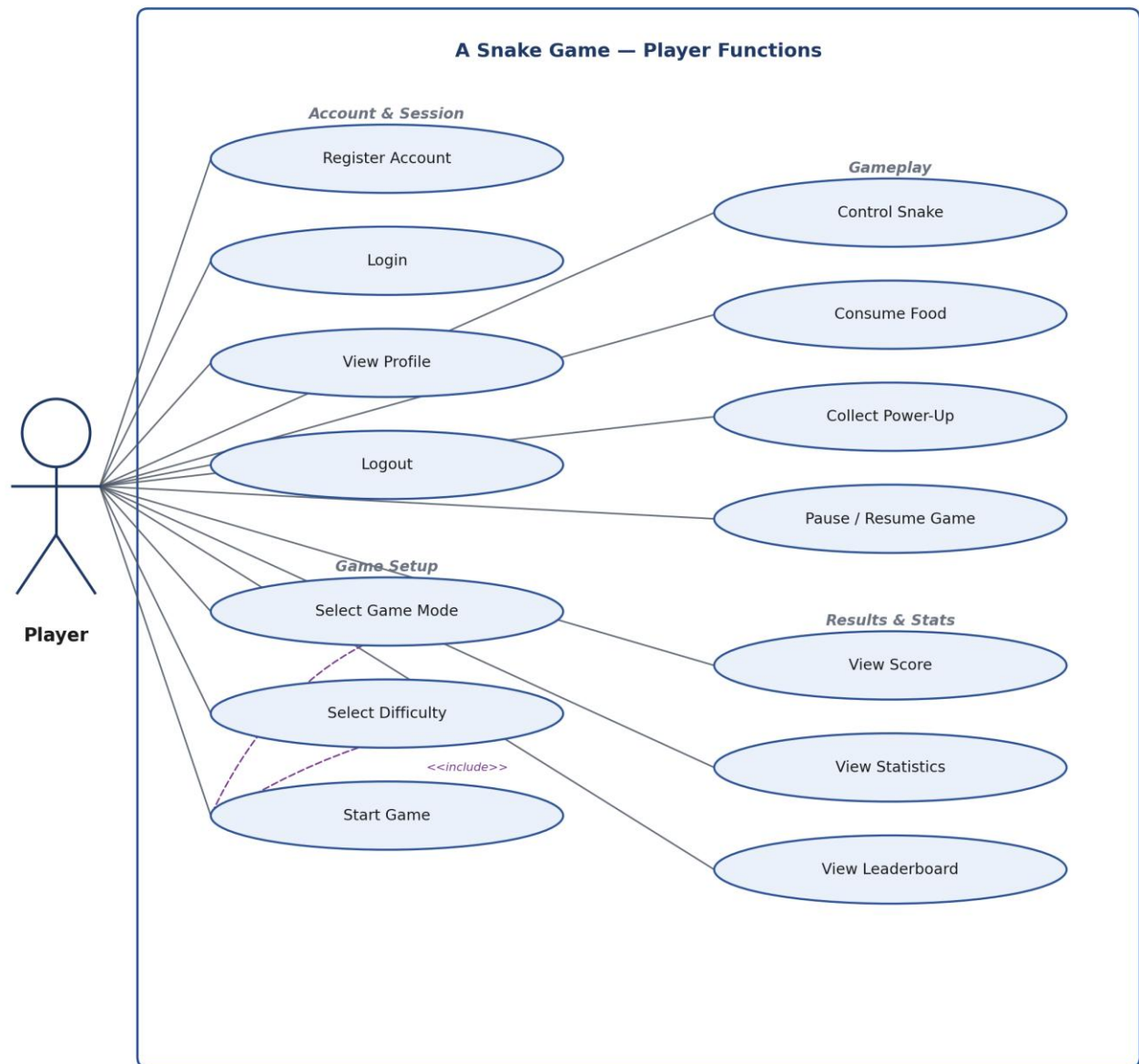


Figure 1: Player Use-Case Diagram — Player actor and associated use cases.

- Register Account
- Login
- View Profile
- Select Game Mode
- Select Difficulty
- Start Game
- Control Snake
- Consume Food
- Collect Power-Up
- Pause/Resume Game
- View Score
- View Statistics

- View Leaderboard
- Logout

The Player actor shall interact with the selected game mode, while the system applies the corresponding gameplay rules.

7.2 Administrator Use-Case Diagram

Actor: Administrator

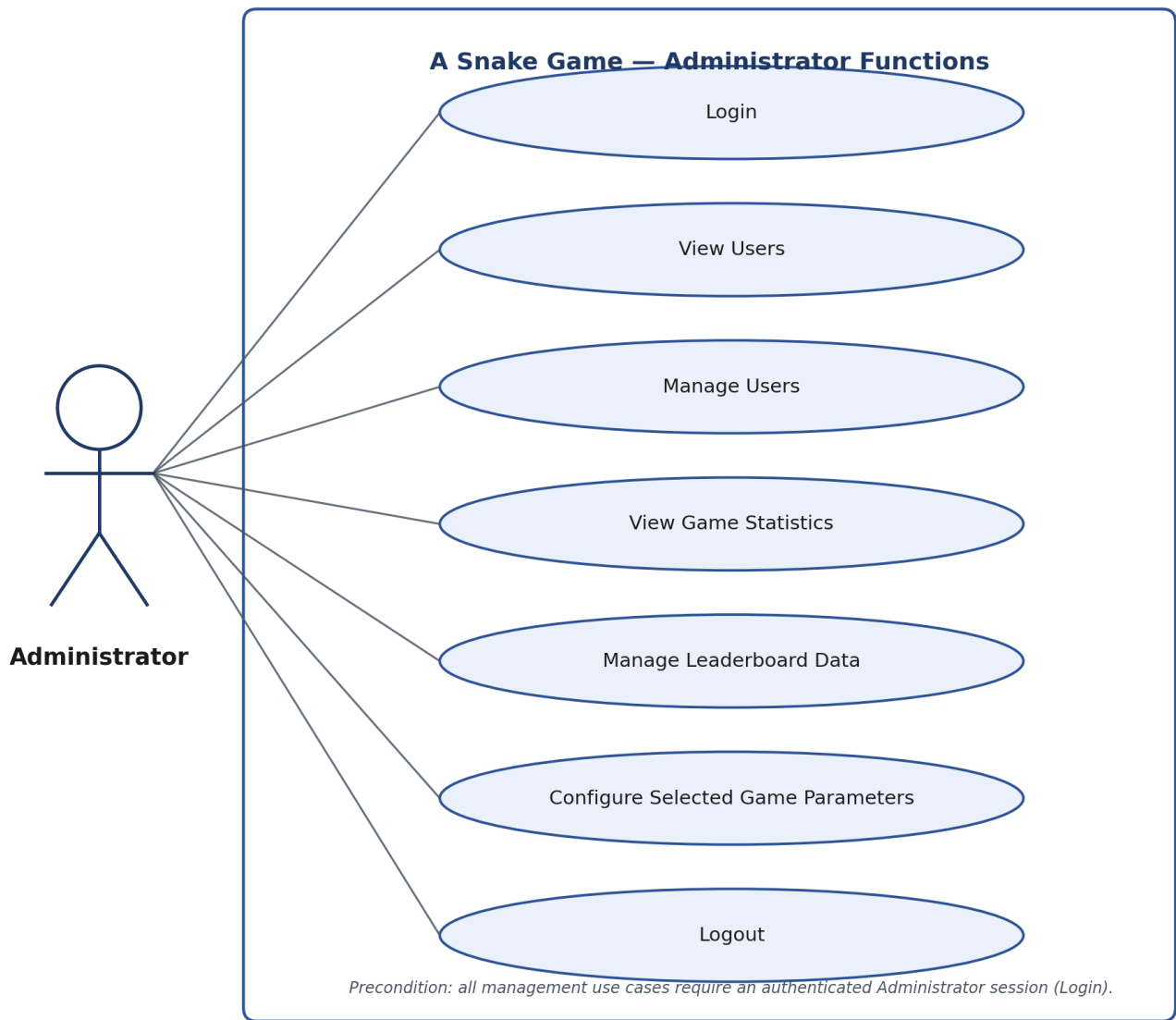


Figure 2: Administrator Use-Case Diagram — Administrator actor and associated use cases.

- Login
- View Users
- Manage Users
- View Game Statistics
- Manage Leaderboard Data

- Configure Selected Game Parameters
- Logout

Administrative use cases shall only be accessible after successful administrator authentication.

8. Requirements Traceability Matrix

Req ID	Requirement / Feature	Module	Test Case	Status
SNAKE-F-001	Player Registration	User Management Module	TC-AUTH-001	N
SNAKE-F-002	Player Authentication	Authentication Module	TC-AUTH-002	N
SNAKE-F-003	Player Profile	Profile Module	TC-PROFILE-001	N
SNAKE-F-004	Logout	Authentication Module	TC-AUTH-003	N
SNAKE-F-005	Start Game	Game Manager	TC-GAME-001	N
SNAKE-F-006	Game Mode Selection	Game Mode Manager	TC-MODE-001	N
SNAKE-F-007	Difficulty Selection	Difficulty Manager	TC-DIFF-001	N
SNAKE-F-008	Pause and Resume	Game State Manager	TC-GAME-002	N
SNAKE-F-009	Snake Movement	Snake Controller	TC-MOVE-001	N
SNAKE-F-010	Food Consumption	Food Manager / Game Engine	TC-FOOD-001	N
SNAKE-F-011	Food Generation	Food Manager	TC-FOOD-002	N
SNAKE-F-012	Collision Detection	Collision Manager	TC-COLLISION-001	N
SNAKE-F-013	Score Calculation	Score Manager	TC-SCORE-001	N
SNAKE-F-014	Obstacle Mode	Obstacle Mode Module	TC-OBSTACLE-001	N
SNAKE-F-015	Power-Up Mode	Power-Up Manager	TC-POWERUP-001	N
SNAKE-F-016	Two-Player Battle	Multiplayer Module	TC-MULTI-001	N
SNAKE-F-017	Survival Mode	Survival/Difficulty Manager	TC-SURVIVAL-001	N
SNAKE-F-018	Classic Mode	Classic Mode Module	TC-CLASSIC-001	N
SNAKE-F-019	Save Game Score	Database / Score Manager	TC-SCORE-002	N
SNAKE-F-020	Leaderboard	Leaderboard Module	TC-LEADER-001	N
SNAKE-NF-001	The system shall process valid play...	Architecture / relevant module	Measured response ≤100 ms	N
SNAKE-NF-002	The game shall maintain at least 30...	Architecture / relevant module	FPS measurement ≥30	N
SNAKE-NF-003	The system shall preserve successful...	Architecture / relevant module	Repeated retrieval matches stored score	N
SNAKE-NF-004	A new player shall be able to start...	Architecture / relevant module	Usability test passes	N
SNAKE-NF-005	The system shall separate major gam...	Architecture / relevant module	Architecture/code review confirms modular separation	N
SNAKE-NF-006	The system shall	Architecture /	Compatibility test	N

	provide consistent...	relevant module	passes	
SNAKE-SR-001	The system shall store passwords us...	Security / relevant module	TC-SEC-001	N
SNAKE-SR-002	The system shall restrict administr...	Security / relevant module	TC-SEC-002	N
SNAKE-SR-003	The system shall validate user-prov...	Security / relevant module	TC-SEC-003	N
SNAKE-SR-004	The system shall prevent normal pla...	Security / relevant module	TC-SEC-004	N
SNAKE-SR-005	The system shall terminate authenti...	Security / relevant module	TC-SEC-005	N

Status codes: N = Not yet implemented; P = Partially implemented; A = Accepted/verified.

Appendix A — Proposed System Modules

- User Management Module
- Authentication Module
- User Interface Module
- Game Engine
- Snake Controller
- Game Mode Manager
- Collision Detection Module
- Food Manager
- Power-Up Manager
- Difficulty Manager
- Score Manager
- Leaderboard Module
- Statistics Module
- Database Module
- Administration Module

Appendix B — Game Mode Summary

Game Mode	Primary Objective	Major Features
Classic	Achieve highest score	Food, growth, collision
Obstacle	Navigate around obstacles	Obstacles, collision detection
Power-Up	Use temporary abilities	Power-ups, effects, durations
Two-Player Battle	Defeat/outscore another player	Two snakes, shared board
Survival	Survive increasingly difficult gameplay	Progressive difficulty, obstacles, speed

Appendix C — Future Enhancements

- Online multiplayer
- AI-controlled snakes
- Custom map creation
- Additional power-ups

- Tournament mode
- Cloud synchronization
- Achievements and badges
- Replay system
- Mobile version
- Additional visual themes